The Laplace–Beltrami Operator on Hurwitz and Hurwitz-like Surfaces

Spectral Gaps, Representation Theory, and Topological Constraints

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Abstract

The Laplace–Beltrami operator on a compact hyperbolic Riemann surface encodes three things simultaneously: the geometry (curvature, metric), the topology (genus, handles), and the symmetry (automorphism group). Its eigenvalues are the natural frequencies of the surface; its eigenspaces are irreducible representations of the symmetry group; and its first cohomology measures the topological obstructions that no amount of geometric deformation can remove. We survey this structure on a family of hyperbolic surfaces arising as quotients of the hyperbolic plane by triangle groups—the Bolza surface (genus 2, $\Delta(2,3,8)$, $\mathrm{GL}(2,\mathbb{F}_3)$), the Klein quartic (genus 3, $\Delta(2,3,7)$, $\mathrm{PSL}(2,7)$), and the Macbeath surface (genus 7, $\Delta(2,3,7)$, $\mathrm{PSL}(2,8)$)—and discuss the tension between maximising symmetry and maximising spectral gap. Companion code computing the cell complexes, homology, and group representations is included.

1. Motivation: From LLMs to Hyperbolic Surfaces

This paper is a companion to *LLMs as Resonant Structures* [1], which develops a spectral theory of large language models in three layers: Green's functions (linear spectral decomposition), the Inverse Scattering Transform (nonlinear spectral decomposition via sparse autoencoders), and sheaf theory (the global obstruction to interpretability).

The sheaf-theoretic layer raises a concrete mathematical question. If one treats the LLM's latent space as a Riemannian manifold M, then local linearisations (SAE decompositions that work in a neighbourhood of input space) form a *presheaf* on M. Polysemanticity—the phenomenon where a single feature means different things in different contexts—is the failure of the gluing axiom: the presheaf is not a sheaf. The first Čech cohomology $H^1(M, \mathcal{F})$ measures the degree of this failure.

The natural next question is: *how much of H^1 is topologically forced?* If the manifold M has nontrivial topology (handles, genus), then Gauss–Bonnet and Hodge theory guarantee that $H^1 \neq 0$ regardless of the metric—no training objective can eliminate it. The Laplace–Beltrami operator Δ_g on M is the bridge between the topology (H^1), the geometry (curvature, holonomy), and the representation theory (how the symmetry group organises the eigenspaces).

I should be honest about what I don't know. My background is in symmetric graphs and symmetric polynomials—algebraic combinatorics, not differential geometry. I view everything through the lens of symmetry. I do not have working intuition for the Laplace–Beltrami operator, and this paper is my attempt to build some.

The surfaces studied here—the Bolza surface, the Klein quartic, the Macbeath surface—were not chosen because they are the right model for an LLM's latent manifold. They were chosen because I already know how to compute with them. In prior work on quantum error-correcting codes [9], I formalised in Lean 4 the fact that a CSS code constructed from a surface tiling encodes exactly

$2g$ logical qubits, where g is the genus. The chain complex condition $\partial^2 = 0$ (which gives the orthogonality of X and Z stabilizers) and the first Betti number $\beta_1 = 2g$ (which counts the logical qubits) are the same H^1 that appears in the sheaf-theoretic picture of polysemanticity. These surfaces are the manifolds for which I have explicit cell complexes, boundary maps, cohomology computations, and representation-theoretic decompositions already in hand.

The important caveat is that optimising for symmetry—which is what makes these surfaces tractable—is *not* the same as optimising for the quantities that matter in the LLM setting. The Laplace–Beltrami training objective proposed by Poole [10]—minimising holonomy to make multi-agent representations locally and globally flat—optimises for *geometric* flatness, not algebraic structure. Symmetry constrains the eigenspaces; flatness constrains the eigenvalues. As we shall see, these pull in opposite directions (Section 6). Nevertheless, highly symmetric surfaces make good starting examples precisely because they are the "spherical cows" of Riemannian spectral theory: simplified enough to compute with, structured enough to reveal the essential trade-offs.

The results suggest that:

- The spectral gap λ_1 controls robustness: how resistant the surface is to low-frequency perturbations (analogous to prompt sensitivity in an LLM).
- Eigenspace multiplicities, forced by irreducible representations of the symmetry group, determine feature granularity (analogous to the resolution of SAE features).
- $H^1 = \mathbb{R}^{2g}$ is a topological invariant: the number of "polysemantic handles" that no amount of training can flatten—and, in the quantum coding setting, the number of logical qubits protected by the code.
- Maximising symmetry (Hurwitz surfaces) and maximising spectral gap (robustness) are in tension—a trade-off between interpretability and stability that may have a direct analogue in LLM design.

2. Triangle Groups and Hyperbolic Surfaces

A triangle group $\Delta(p, q, r) = \langle a, b \mid a^p = b^q = (ab)^r = 1 \rangle$ with $1/p + 1/q + 1/r < 1$ acts on the hyperbolic plane $\mathbb{H}^2$ by isometries, tiling it with hyperbolic triangles of angles π/p, π/q, π/r. If G is a finite group admitting a surjection $\Delta(p, q, r) \twoheadrightarrow G$ with torsion-free kernel $\Gamma \trianglelefteq \Delta$, then the quotient

$$S = \mathbb{H}^2/\Gamma$$

is a compact Riemann surface with $\mathrm{Aut}(S) \supseteq G$ and genus determined by the Riemann–Hurwitz formula:

$$|G| = \frac{4\pi(g-1)}{2\pi\left(1 - \frac{1}{p} - \frac{1}{q} - \frac{1}{r}\right)} = \frac{2(g-1)}{1 - 1/p - 1/q - 1/r}.$$

The *Hurwitz bound* states that $|\mathrm{Aut}(S)| \leq 84(g-1)$ for genus $g \geq 2$, achieved precisely when the triangle group is $\Delta(2, 3, 7)$ (since $1 - 1/2 - 1/3 - 1/7 = 1/42$ gives $|G| = 84(g-1)$). Surfaces achieving this bound are called *Hurwitz surfaces*.

2.1. The family

We study three surfaces, listed with their triangle group data:

Surface	**Genus**	$\Delta(p,q,r)$	$\lvert\mathrm{Aut}\rvert$	**Hurwitz?**	**Group**
Bolza	2	$(2,3,8)$	48	No (max is 84)	$\mathrm{GL}(2,\mathbb{F}_3)$
Klein quartic	3	$(2,3,7)$	168	Yes $(= 84 \cdot 2)$	$\mathrm{PSL}(2,7)$
Macbeath	7	$(2,3,7)$	504	Yes $(= 84 \cdot 6)$	$\mathrm{PSL}(2,8)$

The Bolza surface is the most symmetric genus-2 surface but falls short of the Hurwitz bound: $48 < 84$. The Klein quartic is the smallest Hurwitz surface; the Macbeath surface is the next. Each carries a regular tessellation inherited from the triangle tiling of $\mathbb{H}^2$: the Bolza surface has a $\{3,8\}$ triangulation (16 triangles, 6 vertices of valence 8) dual to a $\{8,3\}$ octagonal tiling, while the Klein quartic has a $\{3,7\}$ triangulation dual to the famous $\{7,3\}$ heptagonal tiling.

3. The Laplace–Beltrami Operator

Equip each surface S with its constant-curvature -1 hyperbolic metric. The Laplace–Beltrami operator Δ_g is defined in local coordinates by

$$\Delta_g f = \frac{1}{\sqrt{|g|}} \sum_{i,j} \partial_i \left(\sqrt{|g|}\, g^{ij}\, \partial_j f \right),$$

where g_{ij} is the metric tensor and g^{ij} its inverse. This generalises the flat Laplacian $\partial^2/\partial x^2 + \partial^2/\partial y^2$ to curved surfaces. On a compact surface without boundary, Δ_g has a discrete spectrum:

$$0 = \lambda_0 < \lambda_1 \le \lambda_2 \le \lambda_3 \le \cdots \to \infty.$$

The eigenfunctions φ_n satisfying $\Delta_g \varphi_n = -\lambda_n \varphi_n$ are the natural vibration modes of the surface—the hyperbolic analogues of $\sin(nx)$ and $\cos(nx)$ on a circle.

3.1. What the spectrum encodes

The spectral gap λ_1. The first nontrivial eigenvalue controls how quickly local perturbations dissipate across the surface. A large spectral gap means the surface is "well-mixed"—perturbations cannot remain localised. A small gap means the surface has near-disconnected regions.

The area. Fixed by Gauss–Bonnet: $\mathrm{Area}(S) = 4\pi(g-1)$. A genus-2 surface has area 4π; genus 7 has area 24π.

The Yang–Yau bound. An upper bound on λ_1:

$$\lambda_1 \le \frac{8\pi(\lfloor g/2 \rfloor + 1)}{\mathrm{Area}(S)} = \frac{2(\lfloor g/2 \rfloor + 1)}{g-1}.$$

For genus 2: $\lambda_1 \le 4$. For genus 3: $\lambda_1 \le 8/3 \approx 2.67$. For genus 7: $\lambda_1 \le 4/3 \approx 1.33$.

The Selberg bound. For arithmetic hyperbolic surfaces (quotients by arithmetic lattices), Selberg conjectured $\lambda_1 \ge 1/4$. The current best result is $\lambda_1 \ge 975/4096 \approx 0.238$ (Kim–Sarnak). For random hyperbolic surfaces of large genus, $\lambda_1 \to 1/4$ almost surely (Brooks–Makover). So $1/4$ is the "generic" spectral gap.

4. The Bolza Surface

The Bolza surface B is defined algebraically by $y^2 = x^5 - x$, with automorphism group $\mathrm{GL}(2, \mathbb{F}_3)$ of order 48. Its fundamental domain is a regular hyperbolic octagon with vertex angles $\pi/4$ (so that $8 \times \pi/4 = 2\pi$ at each vertex). The octagon's sides are identified in pairs according to the standard genus-2 surface word $aba^{-1}b^{-1}cdc^{-1}d^{-1}$.

4.1. The spectrum

The first several eigenvalues, computed numerically by Strohmaier–Uski [2] and Cook [3], are:

λ_n (approx)	Multiplicity	Eigenspace structure
0	1	Trivial representation
3.839	2	2-dimensional irrep of $\mathrm{GL}(2, \mathbb{F}_3)$
5.354	3	3-dimensional irrep
8.250	3	3-dimensional irrep
14.72	2	2-dimensional irrep

The spectral gap $\lambda_1 \approx 3.84$ is conjectured to be maximal among all genus-2 hyperbolic surfaces. The Yang–Yau bound gives $\lambda_1 \leq 4$, so the Bolza surface achieves 96% of the theoretical maximum. This is far above the Selberg baseline of $1/4$: the Bolza surface is a spectral outlier, extraordinarily well-mixed for its genus.

4.2. Representation-theoretic structure

The group $\mathrm{GL}(2, \mathbb{F}_3)$ has order $48 = (9-1)(9-3)$. Its irreducible representations over $\mathbb{R}$ include representations of dimensions 1, 2, 3, and higher. The eigenspaces of Δ_g must be $\mathrm{GL}(2, \mathbb{F}_3)$-representations, so they decompose into direct sums of irreps.

The first nontrivial eigenspace at $\lambda_1 \approx 3.84$ is 2-dimensional: the smallest nontrivial irrep. This means you cannot have a single eigenfunction at λ_1—the symmetry group acts on it and generates a 2-dimensional orbit. The symmetry forces eigenfunctions into multiplets.

4.3. Cohomology and the $(2, 3, 8)$ triangle group

The Bolza surface arises from the exact sequence $1 \to \Gamma \to \Delta(2, 3, 8) \to \mathrm{GL}(2, \mathbb{F}_3) \to 1$ where Γ is a torsion-free Fuchsian group (the fundamental group $\pi_1(B)$). The cell complex inherited from the $(2, 3, 8)$ tiling gives:

- $\{3, 8\}$ triangulation: 16 triangles, 24 edges, 6 vertices (valence 8)
- $\{8, 3\}$ dual: 6 octagons, 24 edges, 16 vertices (valence 3)—the dual graph is the Möbius–Kantor graph
- $\chi = 6 - 24 + 16 = -2$, confirming genus 2

The cellular homology gives Betti numbers $\beta_0 = 1$, $\beta_1 = 4 = 2g$, $\beta_2 = 1$, confirming $H^1(B, \mathbb{R}) \cong \mathbb{R}^4$. The $\mathrm{GL}(2, \mathbb{F}_3)$-action on H^1 decomposes as $V_2 \oplus \overline{V}_2$: a conjugate pair of 2-dimensional irreps (see `bolza-surface/bolza_cohomology.py` for the computation).

5. The Klein Quartic

The Klein quartic is the genus-3 Riemann surface defined by $x^3y + y^3z + z^3x = 0$ in $\mathbb{P}^2$, with automorphism group $\mathrm{PSL}(2,7)$ of order 168. It is the smallest Hurwitz surface: $168 = 84(3-1)$.

5.1. The $(2,3,7)$ triangle group

The Klein quartic arises from $1 \to \Gamma \to \Delta(2,3,7) \to \mathrm{PSL}(2,7) \to 1$. The triangle tiling of $\mathbb{H}^2$ by $(2,3,7)$ triangles descends to:

- $\{3,7\}$ triangulation: 56 triangles, 84 edges, 24 vertices (valence 7)
- $\{7,3\}$ dual: 24 heptagons, 84 edges, 56 vertices (valence 3)
- $\chi = 24 - 84 + 56 = -4$, confirming genus 3

This is the famous heptagonal tiling depicted in countless illustrations of hyperbolic geometry. The 24 heptagons tile the genus-3 surface with 3 meeting at each vertex.

5.2. Representation theory of $\mathrm{PSL}(2,7)$

$\mathrm{PSL}(2,7)$ has 6 conjugacy classes and irreducible representations of dimensions $1, 3, \overline{3}, 6, 7, 8$. The cohomology decomposes as

$$H^1(\mathrm{Klein}, \mathbb{R}) \cong \mathbb{R}^6 \cong V_3 \oplus \overline{V}_3,$$

a conjugate pair of 3-dimensional irreps (see `klein-quartic/klein_quartic_cohomology.py`). This is the representation-theoretic signature of a Hurwitz surface: the cohomology is forced into the smallest nontrivial irreps of the automorphism group.

5.3. Spectral consequences of Hurwitz symmetry

Because $\mathrm{PSL}(2,7)$ has no 2-dimensional irreps, the first nontrivial eigenspace of Δ_g on the Klein quartic must be at least 3-dimensional—the smallest nontrivial irrep available. This is a direct consequence of the Hurwitz bound: maximum symmetry forces the eigenspaces into *larger* multiplets.

Combined with the Yang–Yau bound $\lambda_1 \leq 8/3$ for genus 3, this constrains the spectrum tightly. The Klein quartic cannot have a spectral gap as large as the Bolza surface ($\lambda_1 \leq 2.67$ vs. the Bolza's $\lambda_1 \approx 3.84$), but its eigenspaces are more highly structured.

6. The Macbeath Surface

The Macbeath surface is the genus-7 Hurwitz surface with automorphism group $\mathrm{PSL}(2,8)$ of order $504 = 84 \cdot 6$. It arises from the same $(2,3,7)$ triangle group as the Klein quartic, but with a different (larger) quotient.

6.1. Representation theory of $\mathrm{PSL}(2,8)$

$\mathrm{PSL}(2,8)$ has 9 conjugacy classes with irreducible representations of dimensions $1, 7, 7, 7, 8, 8, 8, 9, 9$. Notably, the smallest nontrivial irrep is 7-dimensional. This forces each nontrivial eigenspace of Δ_g to have dimension at least 7.

The cohomology is $H^1 \cong \mathbb{R}^{14}$, and it decomposes into $\mathrm{PSL}(2,8)$-irreps. The 14-dimensional H^1 provides 14 independent topological obstructions—14 independent directions in which local frames cannot be made globally compatible.

6.2. The spectral gap shrinks

The Yang–Yau bound for genus 7 is $\lambda_1 \leq 4/3 \approx 1.33$. Combined with eigenspaces of dimension ≥ 7, the Macbeath surface has a coarser spectral resolution than either the Bolza or Klein surfaces: fewer distinct eigenvalues in any given range, each with high multiplicity.

7. The Hurwitz Paradox: Symmetry vs. Spectral Gap

7.1. The tension

Comparing the three surfaces reveals a fundamental tension:

Surface	g	$\lvert\mathrm{Aut}\rvert$	Hurwitz?	λ_1 bound	Min eigenspace dim
Bolza	2	48	No	≤ 4	2
Klein	3	168	Yes	≤ 2.67	3
Macbeath	7	504	Yes	≤ 1.33	7

More symmetry means:

- Larger minimum eigenspace dimensions (eigenspaces must be full irreps)
- A lower upper bound on λ_1 (via Yang–Yau, since genus grows with symmetry)
- Coarser spectral resolution (fewer distinct eigenvalues, each highly degenerate)

Less symmetry means:

- Smaller eigenspace dimensions (finer-grained modes)
- A potentially larger spectral gap (the Bolza surface nearly saturates Yang–Yau)
- More distinct eigenvalues in any given spectral range

7.2. Symmetry maximises structure; spectral gap maximises mixing

The Hurwitz bound $84(g-1)$ maximises the number of symmetries a surface of genus g can carry. But maximising symmetry is not the same as maximising the spectral gap. The Bolza surface has less symmetry than the Hurwitz bound allows, but it has the largest spectral gap of any genus-2 surface.

The two criteria optimise different things:

- **Symmetry (Hurwitz)** → the eigenspaces are maximally constrained by representation theory → the spectral structure is most "interpretable" (fewest free parameters, everything determined by group theory).
- **Spectral gap** → perturbations dissipate maximally fast → the surface is most "robust" (hardest to excite nontrivial modes).

These are in tension because achieving the Hurwitz bound requires genus to grow (for larger groups), which pushes the Yang–Yau bound down and forces the spectral gap to shrink.

8. Topological Constraints: Why Curvature Cannot Be Eliminated

8.1. Gauss–Bonnet

On any compact surface of genus $g \geq 2$ with constant curvature $K = -1$:

$$\int_S K \, dA = 2\pi\chi(S) = 4\pi(1-g).$$

The total curvature is a topological invariant. You can redistribute curvature (make some regions flatter and others more curved), but you cannot eliminate it. The negative curvature is topologically forced.

8.2. H^1 and the handles

The first cohomology $H^1(S, \mathbb{R}) \cong \mathbb{R}^{2g}$ records the topological obstructions. Each handle of the surface contributes two independent directions to H^1, and each direction represents a way in which parallel transport around a non-contractible loop can fail to return a vector to its starting orientation.

Surface	Genus	$\dim H^1$	H^1 as Aut-representation
Bolza	2	4	$V_2 \oplus \overline{V}_2$ under $\mathrm{GL}(2, \mathbb{F}_3)$
Klein	3	6	$V_3 \oplus \overline{V}_3$ under $\mathrm{PSL}(2, 7)$
Macbeath	7	14	Irreps of $\mathrm{PSL}(2, 8)$

The pattern is clear: H^1 always decomposes into conjugate pairs of the *smallest* nontrivial irreps of the automorphism group. For the Bolza surface, the smallest is 2-dimensional; for the Klein quartic, 3-dimensional; for Macbeath, the 7-dimensional irreps. The topology forces H^1 to fill out complete irreducible representations—it cannot be "half" of an irrep.

8.3. Holonomy as parallel transport failure

The holonomy group of a Riemannian manifold measures what happens when you parallel-transport a tangent vector around a closed loop. On a flat manifold, the vector returns unchanged (trivial holonomy). On a curved manifold, it returns rotated.

On a genus-g surface with constant curvature -1, the holonomy around the $2g$ generators of $\pi_1(S)$ is nontrivial. These are the "handles" of the surface, and the holonomy around each

handle is determined by the hyperbolic translation length of the corresponding element of the Fuchsian group Γ.

Trivial holonomy would require the surface to be flat—but Gauss–Bonnet forbids this for $g \geq 2$. The holonomy is therefore topologically forced: a consequence of the genus, not a failure of optimisation.

9. Eigenfunctions as Irreducible Representations

9.1. The general principle

If a group G acts on a Riemannian manifold S by isometries, then G acts on each eigenspace $E_\lambda = \{f : \Delta_g f = -\lambda f\}$. By Schur's lemma, E_λ decomposes into a direct sum of irreducible G-representations:

$$E_\lambda \cong \bigoplus_i V_{\rho_i}^{\oplus m_i}.$$

The multiplicities m_i are the Fourier coefficients of the spectral decomposition with respect to the representation ring of G.

This means:

- Each eigenvalue "carries" one or more irreps of the symmetry group.
- The minimum eigenspace dimension equals the dimension of the smallest irrep that appears at that eigenvalue.
- Two eigenvalues can only be equal if their eigenspaces merge into a single (larger) G-representation.

9.2. Consequences for spectral resolution

A group with large minimum irrep dimension forces coarse spectral resolution. $\mathrm{PSL}(2,8)$ has minimum nontrivial irrep dimension 7, so the Macbeath surface cannot have an eigenspace smaller than 7-dimensional (apart from the trivial eigenvalue $\lambda_0 = 0$). In contrast, $\mathrm{GL}(2,\mathbb{F}_3)$ has a 2-dimensional irrep, so the Bolza surface can have eigenspaces as small as 2-dimensional.

This is a direct trade-off: more symmetry buys you more algebraic structure (the eigenspaces are determined by representation theory), but at the cost of spectral resolution (each eigenvalue accounts for more dimensions). The "finest-grained" spectral analysis is available on the least symmetric surfaces.

10. Conclusion

The Laplace–Beltrami operator on a compact hyperbolic surface is the nexus where topology, geometry, and representation theory meet:

Property	What it determines
λ_1 (spectral gap)	Mixing rate; robustness to perturbation
Eigenspace dimensions	Forced by irreps of $\mathrm{Aut}(S)$
$H^1 \cong \mathbb{R}^{2g}$	Topological obstructions (handles)
H^1 as Aut-rep	How symmetry organises the obstructions
Yang–Yau bound	Maximum possible spectral gap for given genus
Selberg bound (1/4)	Generic large-genus baseline
Holonomy	Parallel transport failure around handles

The Hurwitz paradox—that maximising symmetry and maximising spectral gap are in tension—reveals that these surfaces optimise different criteria. Hurwitz surfaces maximise algebraic structure (the representation theory determines everything). Non-Hurwitz surfaces like the Bolza can maximise spectral gap (robustness to perturbation) at the cost of less algebraic constraint.

The topological invariant $H^1 = \mathbb{R}^{2g}$, decomposed into irreducible representations of the automorphism group, is the fundamental datum: it measures what cannot be changed by any geometric deformation, only by changing the topology itself. On a genus-g surface, there are exactly $2g$ independent directions in which curvature is topologically forced and holonomy is necessarily nontrivial. This is a theorem, not an approximation.

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